

# **PROJECT PROPOSAL FOR RESEARCH PROJECT**

**IIT 474-6**

## **AI-Based Tomato Ripening Stage Detection and Sorting System with Shelf-Life Estimation for the Commonly Consumed *Padma* Variety in Sri Lanka Using Computer Vision and Conveyor Automation**

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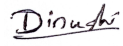
**November 2025**

## Declaration

We, hereby, declare that the research proposal titled

**“AI-Based Tomato Ripening Stage Detection and Sorting System with Shelf-Life Estimation for the Commonly Consumed *Padma* Variety in Sri Lanka Using Computer Vision and Conveyor Automation”** is our original work and has been conducted solely for the purpose of fulfilling the partial fulfilment of the requirement of the course unit **IIT 474-6**. We affirm that this work is free from any form of plagiarism and has not been submitted, in part or in full, for any other course, program, or qualification. All sources used in this proposal have been properly cited and acknowledged.

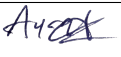
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## List of Abbreviations

|                                                             |         |
|-------------------------------------------------------------|---------|
| Artificial Intelligence                                     | AI      |
| Machine Learning                                            | ML      |
| Convolutional Neural Network                                | CNN     |
| You Only Look Once version 8                                | YOLOv8  |
| Mean Average Precision                                      | mAP     |
| Frames Per Second                                           | FPS     |
| Graphics Processing Unit                                    | GPU     |
| Key Performance Indicator                                   | KPI     |
| Internet of Things                                          | IoT     |
| Pulse Width Modulation                                      | PWM     |
| Powerful Micro controller Board                             | ESP32   |
| Most widely cultivated and commonly consumed tomato variety | ‘Padma’ |

# 1 Introduction

## 1.1 Project Background

Tomato (*Solanum lycopersicum* L.) is one of the most economically important vegetable crops in Sri Lanka. The variety called *Padma* is widely grown in Sri Lanka because it tolerates tropical climate, gives good yield, and is accepted in both retail and wholesale channels. The stage of ripeness at harvest strongly affects flavour, firmness, shelf life, and selling price (Kumar et al. 2023). After harvest, tomatoes are typically graded by hand based on colour and visible surface condition. This manual process is subjective, slow, and depends on the experience and attention level of the worker. The FAO has reported that post-harvest losses in sorting and packaging can reach up to 20% in developing countries (FAO 2022).

With modern computer vision and deep learning, it is now possible to classify fruits, detect defects, and even control sorting hardware in real-time (Li & Zhao 2022). Convolutional Neural Networks (CNNs) and YOLO-style object detectors can recognise fruit maturity stages more reliably than simple colour-threshold methods (Rahman et al. 2021). A recent study by Moya et al. (2025) built a conveyor-integrated tomato grading prototype using YOLOv8. Their system detected tomatoes on a belt, classified them into three classes (green, red, and damaged), and also estimated size. It operated in real-time with classification accuracy of about 99.6% and showed that automated post-harvest grading is practical for industry (Moya et al. 2025). This work is the key baseline of our project.

Separately, shelf-life studies have shown that tomatoes harvested at different ripeness stages survive for different numbers of days before becoming unsellable. For example, controlled experiments link firmness, colour change and visible surface condition over time to remaining market life, and also attempt to model spoilage progression using physical or sensor measurements (Naim et al. 2024, Tarlak & Akbas 2023). However, most of these studies are based on laboratory conditions and do not connect real-time visual grading with automatic decision-making during sorting.

The existing work therefore suggests two opportunities: (i) adapting automated grading and sorting to local tomato varieties such as *Padma*, and (ii) using visually detected ripeness stage to estimate remaining shelf life in a way that can directly support packaging and dispatch decisions in Sri Lankan post-harvest practice.

## 1.2 Problem Definition

This project addresses four core problems:

1. **Real-time ripeness classification for Padma tomatoes:** How can AI-based image processing accurately identify four ripeness stages of the *Padma* tomato in real time on a moving conveyor system?
2. **Defect detection and sorting automation:** How can defective tomatoes (e.g., bruised, cracked or rotted) be identified and physically separated using a mechanical diverter?
3. **Shelf-life prediction:** How can the system estimate the remaining shelf life of each tomato, based on its visually detected ripeness stage and observed quality, so that packing and selling decisions can be prioritised? Prior work on shelf-life modelling often depends on controlled sensors (temperature, humidity, firmness, microbiological growth curves) and is not integrated with on-belt visual detection or local varieties (Naim et al. 2024, Tarlak & Akbas 2023).
4. **Operational decision support:** How can we collect and visualize useful Key Performance Indicators (KPIs) such as throughput rate, class distribution, defect ratio, and predicted shelf life to assist in operational and marketing decisions?

Existing work shows that CNN and YOLO-based approaches can grade or sort tomatoes with high accuracy (Rahman et al. 2021, Li & Zhao 2022, Moya et al. 2025). Huang et al. (2025) further improved YOLO-based tomato ripeness detection using multi-scale feature fusion and a tiny-object detection head, achieving strong performance even when tomatoes overlap in greenhouse scenes (Huang et al. 2025). But most published systems:

- were trained on non-local tomato varieties,
- used only 2–3 classes of ripeness,
- did not attempt shelf-life prediction linked to visual grade on a live conveyor.

There is very limited work targeting Sri Lankan post-harvest practice, with local lighting, local handling conditions, and a locally important variety.

### 1.3 Project Aim and Objectives

**Aim:** To design and develop an AI-based real-time system to detect, classify, and sort *Padma* tomatoes based on ripeness stage and visible defects, and to estimate remaining shelf life.

**Specific Objectives:**

- To collect and label a local dataset of more than 5500 *Padma* tomato images covering four ripeness stages and one defect class under real farm and post-harvest lighting conditions.
- To develop and train a Deep Learning model that performs multi-class classification and defect detection.
- To design and construct a mechanical conveyor prototype mechanism that automatically sorts tomatoes according to detected ripeness stage and defects.
- To integrate and evaluate a shelf-life prediction and KPI monitoring system that estimates remaining freshness (in days) and displays class distribution, defect ratio, and throughput on a live dashboard.

## 2 Project Description

The proposed system is an AI-driven tomato grading and sorting prototype that uses computer vision and mechanical actuation. The system:

- classifies each tomato into one of four maturity stages,
- flags tomatoes with external defects (cracks, bruises),
- predicts remaining shelf life,
- physically diverts certain classes off the conveyor,
- logs KPIs for decision-making.

A high-resolution camera (e.g. USB HD camera) is mounted over a motorised conveyor. Each tomato passing under the camera is detected and classified in real time using a deep learning model. A microcontroller (Arduino) then triggers an actuator (servo or pneumatic arm) to push according to the ripeness stage or “defective stage” tomatoes off the main line.

### 2.1 Ripeness Stages

We will work with four ripeness stages plus a defect class:

| Stage | Name             | Visual indicators                           |
|-------|------------------|---------------------------------------------|
| 1     | Green / Unripe   | Uniform green colour                        |
| 2     | Breaker          | Green-yellow transition, <30% red area      |
| 3     | Turning-Red      | Orange to pink surface, 30–80% red coverage |
| 4     | Red / Fully Ripe | More than 80% red surface                   |
| –     | Defect           | Cracks, bruising, rot                       |

Table 1: Proposed tomato grading classes for the *Padma* variety.



Figure 1: A batch of tomatoes with different ripeness stages, sizes, and defects included.

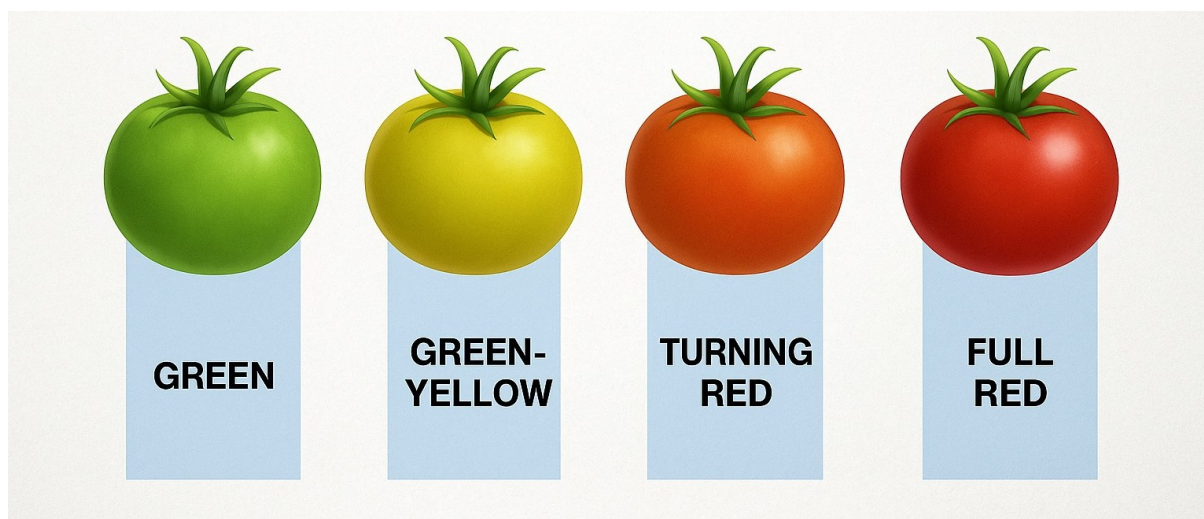


Figure 2: Ripening stages of tomato (Stage 1 to Stage 4).

## 2.2 Real-Time Sorting and Control Flow

The detection output is sent to a microcontroller (Arduino UNO or ESP32). When a tomato is classified to a ripen stage or “defect one”, the microcontroller activates a servo or pneumatic arm to knock that tomato into a side bin. This is similar in spirit to the conveyor-integrated YOLOv8 grading work by Moya et al. (2025), but adapted to:

- four maturity stages instead of only “green” and “red”,
- an explicit defect class,
- locally cultivated *Padma* tomatoes,
- an added shelf-life prediction step.

We will also generate KPIs such as throughput (tomatoes/minute), class distribution, defect ratio, and estimated freshness window.



Figure 3: High-level processing and actuation flow of the proposed system.

## 2.3 Literature Review

Multiple studies confirm that deep learning can classify tomato ripeness with high accuracy:

- Moya et al. (2025) built a YOLOv8 conveyor system that achieved 99.6% classification accuracy and 97.1% size estimation accuracy using three classes (green, red, damaged). This work is the golden baseline we build on.
- Huang et al. (2025) proposed AITP-YOLO, which adds a small-object detection head and multi-scale feature fusion to improve ripeness detection of tomatoes in cluttered greenhouse images. The model also runs efficiently on edge hardware (Jetson).

- Liu et al. (2019) used traditional machine learning (colour features + SVM) to distinguish mature vs immature tomatoes, achieving about 0.9 F1-score, but performance drops under changing light and occlusion.
- Lightweight CNN-based work (Nalupano et al. 2024, Ifmalinda et al. 2023) showed 90–94% accuracy for 3-class maturity levels (unripe / ripe / overripe), but mostly in controlled environments and without any physical sorting system.
- Rahman et al. (2021) reported over 95% accuracy for tomato grading using CNNs; however, that work still relied on static, pre-captured images and not on a moving belt.

Comparisons in the literature show that YOLO-based detectors generally achieve the best trade-off between accuracy and real-time speed and are suitable for industrial conveyor systems. Enhanced YOLO variants further improve robustness for small fruits and overlapping objects (Huang et al. 2025).

| Study / Year            | Technique                            | Focus / Dataset                                                 | Main Results                                          | Limitations                                                |
|-------------------------|--------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------|
| Huang et al. (2025)     | AITP-YOLO (YOLOv10s variant)         | Multi-scale tomato ripeness detection in greenhouse conditions. | 92–93% precision/recall; robust for small fruits.     | Evaluated only in greenhouse lighting.                     |
| Liu et al. (2019)       | HOG + SVM + Colour features          | Mature vs. immature tomato detection using traditional ML.      | F1 $\approx$ 0.9                                      | Sensitive to lighting, occlusion, and background noise.    |
| Nalupano et al. (2024)  | MobileNetV2 / ShuffleNet CNNs        | 3-class (unripe, ripe, overripe) classification.                | $\approx$ 94% accuracy.                               | Low specificity ( $\approx$ 0.64); limited dataset.        |
| Ifmalinda et al. (2023) | HSV / Lab Colour Model (Android app) | Mobile image-based ripeness classification.                     | Strong correlation with chromameter readings.         | Limited adaptability under variable light.                 |
| Moya et al. (2025)      | YOLOv8 + size estimation             | Conveyor-integrated system for grading and sizing.              | 99.6% classification, 97.1% size estimation accuracy. | Only three classes (green, red, damaged); single cultivar. |

Table 2: Summary of Supporting Articles Related to Tomato Ripeness Classification

| Category                       | Typical Accuracy | Deployment Scope          | Key Observations                                                          |
|--------------------------------|------------------|---------------------------|---------------------------------------------------------------------------|
| Classical ML<br>(HOG-SVM, HSV) | 80–92%           | Lab or mobile<br>(manual) | Simple, low-cost, but poor lighting adaptation and low scalability.       |
| CNN (VGG, MobileNet)           | 93–96%           | Controlled indoor         | Good baseline accuracy; lacks real-time integration or hardware link.     |
| YOLO-based detectors           | 97–99%           | Industrial, real-time     | Best trade-off between accuracy and speed; adaptable to sorting lines.    |
| Enhanced YOLO<br>(AITP-YOLO)   | 92–93% mAP       | Greenhouse or field edge  | Robust for small or overlapping fruits; lightweight for embedded devices. |

Table 3: Comparative Analysis of Tomato Classification Techniques

| Study / Year          | Technique / Model                            | Focus / Dataset                                          | Main Findings                                                               | Limitations                                                        |
|-----------------------|----------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------|
| Naim et al. (2024)    | IoT + ML (Random Forest, SVR, KNN)           | Tomato sensor data (temp, humidity, pH, colour, weight). | RF reached $R^2=0.999$ for shelf-life prediction using physical parameters. | Lab-style indoor sensor setup; no real conveyor or sorting.        |
| Tarlak & Akbas (2023) | Predictive microbiology (Gompertz / Baranyi) | Spoilage rate modelling under different temps.           | Growth curves model spoilage progression over time.                         | Focus on microbial spoilage models, not visual maturity on a belt. |

Table 4: Summary of Supporting Articles Related to Tomato Shelf-Life Prediction

## 2.4 Identified Research Gaps

From the literature:

- **Limited class granularity:** Most systems use only 2–3 ripeness classes. Our system uses **4 ripeness stages + 1 defect class**, which is closer to what farmers and buyers actually use.
- **Variety consideration:** Most existing studies do not include the commonly cultivated *Padma* variety. Although *Padma* Variety is widely grown and consumed across the country; however, its shelf life has not yet been predicted in existing research.
- **Shelf-life prediction is not integrated with grading:** Existing systems like Moya et al. (2025) sort tomatoes on a conveyor, but they do not predict **remaining shelf life** according to ripeness stage. Shelf-life studies such as Naim et al. (2024) either use sensors and lab measurements, or track storage over days, but they are not linked to visual ripeness classes in real time.

- **Real market conditions are missing:** Most shelf-life work is done in controlled conditions (fixed temperature, humidity, lab containers). There is almost no work predicting how long a tomato will remain sellable in normal post-harvest conditions. We address this by keeping samples from each ripeness stage at normal conditions and tracking how many days they remain acceptable.
- **No unified decision layer:** KPI dashboards and decision analytics (throughput, defect ratio, shelf-life window) are rarely included. We explicitly log throughput, defect ratio, class distribution, and estimated remaining shelf life to support harvesting and packing decisions.

## 2.5 Novelty of This Project

The novelty of this project is:

- The dataset will be collected from the **Regional Agriculture Research & Development Center, Bandarawela**, covering the *Padma* tomato variety with four ripeness stages and defects. Each stage is expected to have **over 1,500 images**.
- A physical conveyor-based sorting prototype inspired by Moya et al. (2025) but adapted to Sri Lankan conditions. The system will be extended to sort tomatoes not only by ripeness but also by **three main size categories** after ripeness grading.
- **A shelf-life prediction component** that estimates how long each tomato can still be sold or consumed, based on ripeness stage within the normal condition, and that links this to a live KPI dashboard for users. This will be based on observing *Padma* tomatoes over time at normal ambient conditions, not in a controlled environment .

### 3 Methodology

The methodology follows a design-and-implementation pipeline with six main stages: data acquisition, dataset preparation, model development, hardware integration, software design, and testing/evaluation. The approach is experimental and iterative: we measure performance, refine, and retest.

#### 3.1 Overall Approach

The system will (i) capture tomato images on a conveyor, (ii) classify ripeness stage and defect status using a deep learning model, (iii) estimate shelf life, (iv) trigger an actuator to sort tomatoes, and (v) log KPIs to support decision-making. This joins computer vision, Deep Learning and mechatronics into one loop. This pipeline is similar in spirit to the conveyor-integrated YOLOv8 system of Moya et al. (2025), but extended to four ripeness stages for *Padma* and with a shelf-life prediction component.

The project will use quantitative metrics such as model accuracy, actuator response time, and throughput (tomatoes per minute). We will also evaluate how stable the model is under different lighting, belt speeds, and tomato orientations.

#### 3.2 Dataset Preparation

##### *Image Collection*

We plan to capture approximately more than 5,500 images of *Padma* tomatoes using consistent LED lighting to avoid major colour distortion. Multiple camera angles and backgrounds will be used to improve robustness.

##### *Annotation*

Each tomato instance will be annotated with:

- bounding box,
- ripeness stage (1–4),
- defect / non-defect.

Annotation will be done using tools such as LabelImg or Roboflow.

### *Augmentation*

To avoid overfitting, we will apply data augmentation:

- rotation and flips,
- brightness / saturation variation,

### *Dataset Split*

Data will be split into:

- Train: 70%
- Validation: 20%
- Test: 10%

We will enforce class balance so that the model does not bias heavily toward Stage 4 (fully ripe).

## **3.3 Model Development**

### *Architecture Selection*

We will evaluate:

- Deep learning model for real-time detection + classification directly on full frames.

Most accurate deep learning model is expected to be the main model because of its high mean Average Precision (mAP) and single-shot detection capability for real-time conveyor environments (Moya et al. 2025).

### *Training Process*

Training will be done in Python. Planned steps:

- Optimiser: Eg:Adam (tuned learning rate).
- Loss: cross-entropy for class prediction; bounding-box regression loss for localisation.
- Early stopping: stop when validation accuracy stops improving to avoid overfitting.

Model quality will be reported using Precision, Recall, F1-score, Accuracy, and mAP.

### *Ripeness and Defects Detection Module*

A model will identify ripeness stage and bruises, cracks and rotted once. This lets the system make sorting decisions based on both ripeness stage and apparent quality.

### **3.4 Shelf-Life Prediction**

We will run a controlled post-harvest experiment: tomatoes of known stage will be harvested and visually monitored over several days. We will record when each tomato becomes unsellable (Eg: too soft, fungal/rotted, severe blemish). Using these observations, we will estimate the shelf life according to the experimental results of each ripeness stage *remaining shelf life in days* from the visual cues at capture time. That prediction will be displayed along with the class label.

### **3.5 Hardware Integration**

Hardware components:

- **Computing unit:** Laptop with NVIDIA Graphics.
- **Conveyor system:** DC motor belt with PWM speed control.
- **Actuation:** Servo motor or pneumatic arm controlled via Arduino/ESP32.
- **Lighting:** LED strip with diffuser to stabilise visual conditions.
- **Sensing:** Proximity / IR sensor to detect tomato arrival under the camera and sync capture.

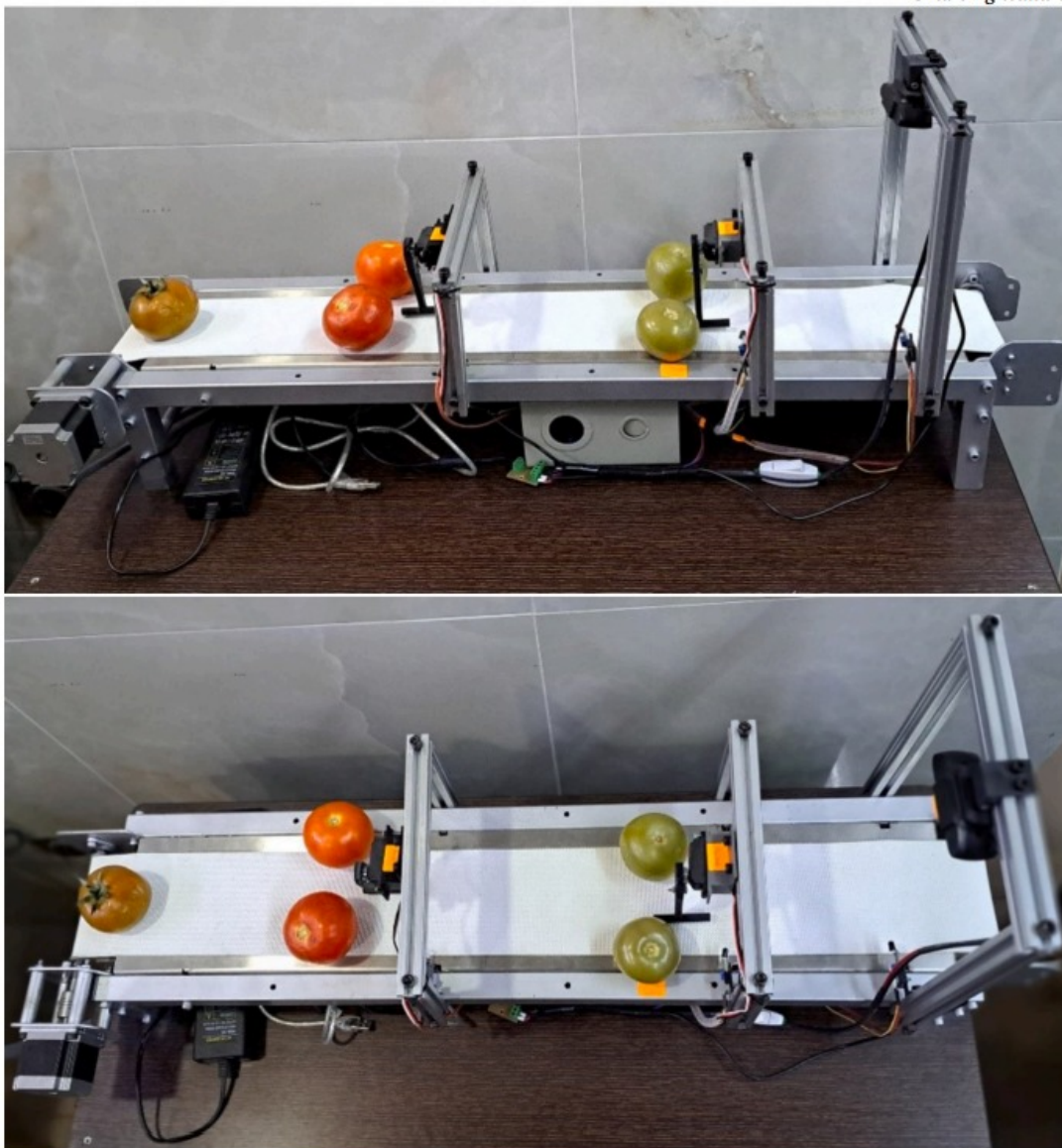


Figure 4: Existing YOLOv8-based conveyor prototype from prior research, which inspires the proposed design from (2025) Moya et al. (2025).

The control flow is:

1. Camera captures frame.
2. Deep Learning model classifies ripeness stage and defect status.
3. If “Stage 4” or “Defect”, send signal to Arduino/ESP32.
4. Arduino triggers servo/actuator to divert tomato.
5. According to ripeness level predict the shlef life and system logs event + KPI.

### 3.6 Software Design

#### *Pipeline*

- Frame acquisition.
- Pre-processing (resize, normalise).
- Model inference (ripeness stage + defect).
- Shelf-life estimation model (regression output in days).
- Action signal to actuator controller.
- KPI logging into a local SQLite database.
- Live dashboard to show counts, class distribution, defect ratio, throughput.

#### *KPI Analytics*

Example KPIs:

- Detection Rate = Detected Tomatoes / Total Tomatoes.
- Defect Ratio = Defective Tomatoes / Total Tomatoes.
- Sorting Throughput = Tomatoes sorted per minute.
- Yield Distribution = (Stage 1 + Stage 2 + Stage 3) / Total Tomatoes.
- Estimated Remaining Shelf Life (average, per batch).

### 3.7 Testing and Evaluation

We will evaluate the system in four phases:

1. **Control Environment tests:** Controlled lighting, constant conveyor speed. Tune camera angle and lighting.
2. **Stress tests:** Vary lighting, belt speed, fruit orientation.
3. **Field validation:** Run pilot trial in a local pack-house using real *Padma* tomatoes.
4. **Benchmarking:** Compare AI sorting vs. manual sorting time and consistency.

## 4 Resources Needed

### Hardware

- Camera module (HD USB camera).
- Conveyor belt with Stepper motor + motor Controller.
- Servo motor or pneumatic actuator.
- Arduino microcontroller.
- LED lighting frame with diffuser.
- Storage crates for shelf-life tracking.

### Software

- Python,PyTorch for model training.
- Deep learning model framework for detection.
- Flask (or Django) for live dashboard.
- SQLite for KPI logging.

### Other Tools

- Wooden/steel frame for the diverter mechanism.
- Multimeter and basic electronics tools for wiring.
- Local supply of *Padma* tomatoes from farms/markets for data collection. (From tomato collectors related to the Regional Agriculture Research & Development Center,Bandarawela)

## Resource Table

| Category    | Items / Tools                                           | Purpose                                               |
|-------------|---------------------------------------------------------|-------------------------------------------------------|
| Hardware    | Conveyor belt, servo/pneumatic arm, LED lighting        | Real-time capture and physical sorting                |
| Software    | Python, deep learning model, Flask, SQLite              | Model Creation, AI inference, dashboard, data logging |
| Dataset     | Locally captured <i>Padma</i> images                    | Training and validation                               |
| Other Tools | Arm, wooden/steel frame, Arduino IDE, multimeter, wires | Prototype construction and testing                    |

Table 5: Summary of required resources.

## 5 Individual Contribution and Work Plan

| Enrolment No.  | Name                  | Key Responsibilities                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UWU/IIT/21/017 | K.R.D.H. Gunawardhana | <ul style="list-style-type: none"> <li>• Participate in data collection and labeling of the local Padma tomato dataset.</li> <li>• Lead shelf-life prediction system development and KPI monitoring integration.</li> <li>• Support prototype development and testing.</li> <li>• Contribute to documentation.</li> </ul>                                                                                                                                  |
| UWU/IIT/21/019 | P.L.H. Hirushan       | <ul style="list-style-type: none"> <li>• Participate in data collection and labeling of the local Padma tomato dataset.</li> <li>• Lead the design and construction of the mechanical conveyor prototype.</li> <li>• Implement the diverter mechanism for sorting.</li> <li>• Support hardware integration and system evaluation.</li> <li>• Contribute to documentation.</li> </ul>                                                                       |
| UWU/IIT/21/062 | W.K. Gayan            | <ul style="list-style-type: none"> <li>• Participate in data collection and labeling of the local Padma tomato dataset.</li> <li>• Lead the development and training of the Deep Learning model for ripeness and defect detection.</li> <li>• Integrate the model with hardware for real-time inference.</li> <li>• Support system testing and performance optimization.</li> <li>• Contribute to documentation and model evaluation reporting.</li> </ul> |

## 6 Project Timeline

The expected timeline along which the research is planned to proceed.

### **Month 1–4 (Nov 2025 – Feb 2026): Literature Review and Background Study**

- Review vision-based tomato grading, Deep learning model based conveyor systems, and sorting automation.
- Finalise problem scope for the Padma tomato variety.
- Define dataset requirements and hardware constraints.

### **Month 1–6 (Nov 2025 – Apr 2026): Dataset Collection and Preparation**

- Capture Padma tomato images in real conditions (multiple ripeness stages and defects).
- Label each tomato with ripeness stage (1–4) and defect / non-defect.
- Augment data and organise train / validation / test splits.

### **Month 4–6 (Feb 2026 – Apr 2026): Model Training and Validation**

- Train the Deep learning model for multi-class ripeness and defect detection.
- Tune accuracy, precision, recall, and F1-score.
- Test real-time inference on live video / conveyor footage.

### **Month 5–8 (Mar 2026 – Jun 2026): Conveyor Design and Prototype Assembly**

- Build the conveyor-based sorting rig with camera mount, lighting frame, and sensors.
- Implement actuator control (servo / pneumatic) via Arduino .
- Prepare physical housing and safety enclosure for operation.

### **Month 6–10 (Apr 2026 – Aug 2026): System Integration and Testing**

- Deploy the trained model to the computer.
- Connect inference output to actuator control for physical removal of tomatoes (Stage 4 / Defect).

- Debug the full loop: camera → AI model → actuator → logging.

#### **Month 7–10 (May 2026 – Aug 2026): KPI Evaluation and Performance Analysis**

- Measure throughput (tomatoes/minute), defect ratio, and classification accuracy.
- Record shelf-life predictions and compare with observed shelf-life tracking.
- Benchmark AI sorting vs manual grading in terms of speed and consistency.

#### **Month 8–10 (Jun 2026 – Aug 2026): Documentation and Final Submission**

- Prepare final report, figures, and presentation materials.
- Summarise KPI dashboard results and system performance.
- Deliver final demonstration of the working prototype.

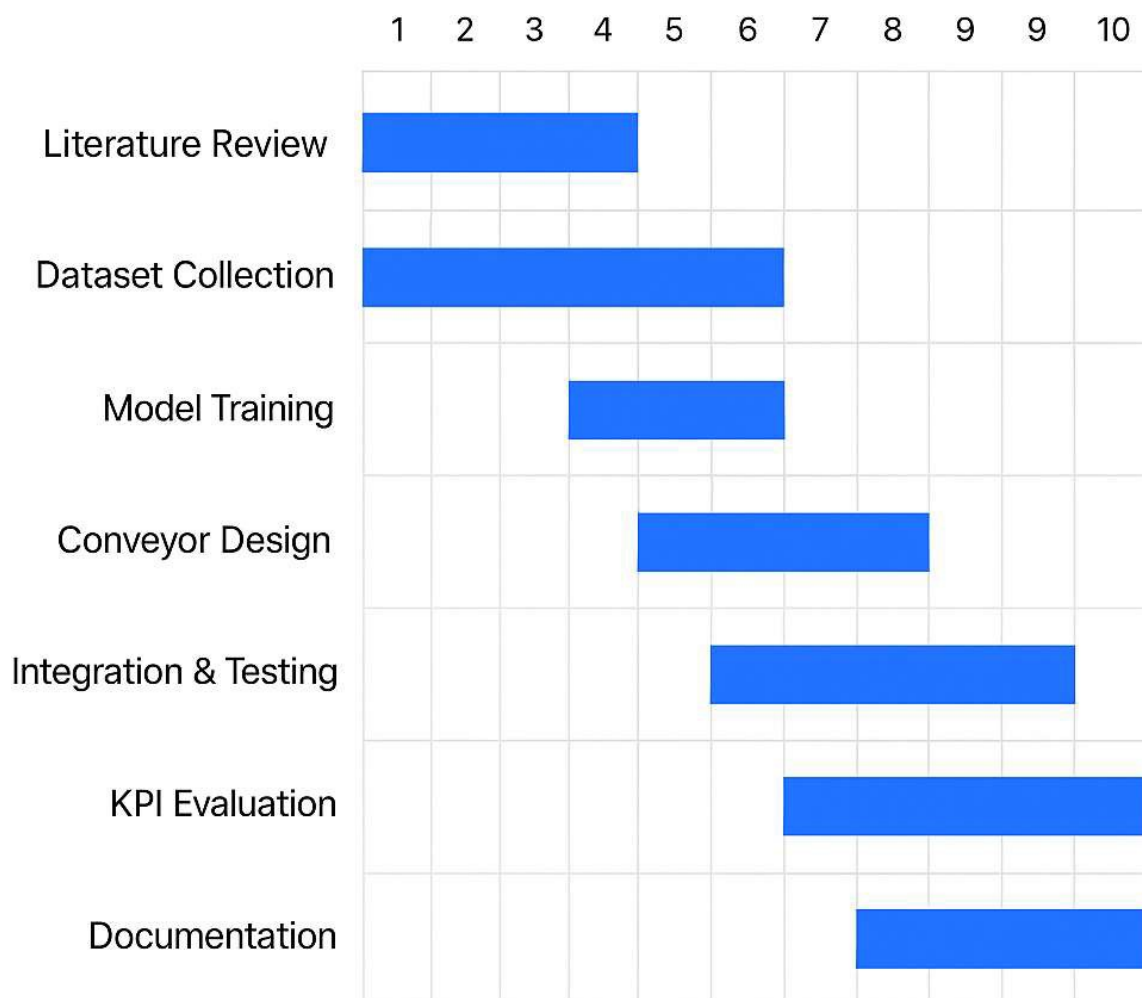


Figure 5: Gantt-style timeline of project phases (Nov 2025 – Aug 2026).

## Project Milestones

Table 6: Project Milestones and Assigned Responsibilities

| No. | Milestone / Task                                                                             | Timeframe           | Responsible Member(s)                                     |
|-----|----------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------|
| M1  | Literature review and background study. Define project scope, objectives, and methodology.   | Nov 2025 – Feb 2026 | UWU/IIT/21/017<br>UWU/IIT/21/019 (Lead)<br>UWU/IIT/21/062 |
| M2  | Data collection: capture and label Padma tomato images for four ripeness stages and defects. | Nov 2025 – Apr 2026 | UWU/IIT/21/017<br>UWU/IIT/21/019<br>UWU/IIT/21/062 (Lead) |
| M3  | Dataset annotation, verification, and augmentation for model training.                       | Jan 2026 – Apr 2026 | UWU/IIT/21/017 (Lead)<br>UWU/IIT/21/019<br>UWU/IIT/21/062 |
| M4  | Deep learning model design, training, and optimization.                                      | Feb 2026 – Apr 2026 | UWU/IIT/21/019 (Lead)<br>UWU/IIT/21/017<br>UWU/IIT/21/062 |
| M5  | Conveyor prototype design, actuator and sensor integration.                                  | Mar 2026 – Jun 2026 | UWU/IIT/21/062 (Lead)<br>UWU/IIT/21/017<br>UWU/IIT/21/019 |
| M6  | Shelf-life experiment setup, data recording, and KPI analysis.                               | Apr 2026 – Aug 2026 | UWU/IIT/21/017<br>UWU/IIT/21/019 (Lead)<br>UWU/IIT/21/062 |
| M7  | System integration and performance evaluation (field testing, calibration).                  | May 2026 – Aug 2026 | UWU/IIT/21/017<br>UWU/IIT/21/019<br>UWU/IIT/21/062 (Lead) |
| M8  | Documentation, report preparation, and proposal refinement.                                  | Jun 2026 – Aug 2026 | UWU/IIT/21/017 (Lead)<br>UWU/IIT/21/019<br>UWU/IIT/21/062 |

## 7 Ethical and Environmental Considerations

This project uses only images of tomatoes. No personal data or human identity data will be collected. All data will be stored securely and accessed only by the research group and supervisors.

Image capture is non-destructive. We do not intentionally damage tomatoes just to collect data. We aim to minimise food waste. Tomatoes used in experiments will either (i) enter normal consumption/processing if still good, or (ii) be disposed of properly if spoiled.

Electrical and mechanical parts (motors, actuators, lighting) will be operated within safe limits to avoid overheating or unsafe lab conditions. Low-power hardware will be preferred where possible to reduce energy use.

The project supports sustainable agriculture by:

- reducing post-harvest loss through more consistent grading,
- Predict the shelf life of the tomatoes early,
- improving sorting speed with less manual fatigue.

## 8 Conclusion

This proposal presents a real-time tomato grading and sorting system designed specifically for the *Padma* variety. The system integrates computer vision, deep learning, embedded hardware, and mechatronic actuation to achieve automated, data-driven post-harvest management.

Building upon the conveyor-based grading approach of Moya et al. (2025), which demonstrated that on-belt tomato inspection and sorting is feasible at industrial speeds, our work extends the concept with four ripeness stages (beyond the conventional “green” and “red”), an explicit “defect” class, locally collected *Padma* data, and a novel shelf-life estimation module.

In this module, a controlled post-harvest experiment will be conducted where tomatoes of known ripeness stages are harvested and visually monitored over several days. By recording the point at which each tomato becomes unsellable. We will empirically estimate the remaining shelf life for each ripeness stage in days. These predictions will be integrated into the system’s output, displayed alongside the ripeness classification results. A KPI dashboard will further visualize system performance and provide real-time operational insights.

By completing this project, we aim to:

- reduce manual grading errors and labour dependency,
- improve quality consistency and market readiness,
- enable data-driven shelf-life prediction to support supply chain decisions,
- provide users and pack-houses with decision-support analytics.

Overall, this project delivers a deployable, locally relevant prototype for smart agriculture in Sri Lanka. It represents a practical step toward scalable, automated, and data-informed post-harvest handling that supports quality assurance and reduces food waste.

## References

- FAO (2022), 'Reducing post-harvest losses in fresh produce', Food and Agriculture Organization of the United Nations. Accessed 10 October 2025.
- Huang, W., Liao, Y., Wang, P., Chen, Z., Yang, Z., Xu, L. & Mu, J. (2025), 'Aitp-yolo: Improved tomato ripeness detection model based on multiple strategies', *Frontiers in Plant Science* **16**, 1596739.
- Ifmalinda, Andasuryani & Rasinta, I. (2023), Identification of tomato ripeness levels (*lycopersicon esculentum* miil) using android-based digital image processing, in 'IOP Conference Series: Earth and Environmental Science', Vol. 1182, p. 012003.
- Kumar, S., Reddy, P. & Patel, R. (2023), 'Tomato maturity detection using deep learning and iot for smart agriculture', *Computers and Electronics in Agriculture* **213**, 108229.
- Li, Y. & Zhao, J. (2022), 'Deep learning-based fruit ripeness detection: A review', *Postharvest Biology and Technology* **193**, 112029.
- Liu, G., Mao, S. & Kim, J. H. (2019), 'A mature-tomato detection algorithm using machine learning and color analysis', *Sensors* **19**(9), 2023.
- Moya, V., Guerra, M., Pazmiño, K., Abedrabbo, F., Chicaiza, F. A. & Pozo-Espín, D. (2025), 'Tomato classification with yolov8: Enhancing automated sorting and quality assessment', *Smart Agricultural Technology* **12**, 101221.
- Naim, S., Priyanka, D. & Ashish, K. (2024), 'Tomato shelf life prediction using iot and machine learning', *International Journal of Advanced Research in Computer Science and Software Engineering* **14**(9), 45–52.
- Nalupano, J. M. G., Omagap, M. P., Fortaleza, K. J. G., Ecraela, F. R. B. & Orquia, J. J. D. (2024), 'Classification of kinalabasa tomato using convolutional neural network', *Journal of Innovative Technology Convergence* **6**(3), 99–110.
- Rahman, M., Hossain, T. & Ferdous, Z. (2021), 'Automated tomato grading using convolutional neural networks', *Journal of Agricultural Informatics* **12**(1), 43–56.

Tarlak, F. & Akbas, M. (2023), 'Predictive modelling of food spoilage using gompertz and baranyi models', *Foods* **12**(4461), 1–15.